

Novelty Analysis

Analysis of Novelty, Transience, and Resonance

PENPAL Project

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0.1 Summary

- Humans produce higher novelty (surprise) and higher transience than the model, indicating stronger deviations from earlier story context.
- Both agents display a positive novelty–resonance relationship (innovation bias), but the slope is steeper for humans.
- This indicates that surprising human turns exert greater forward influence on narrative development.
- user2 contributions show lower variance and lower informational impact relative to human inputs.
- Humans are the primary drivers of narrative innovation and thematic redirection.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
# Load Field Data
df_res_field <- read_csv(here("human-human/data/processed/novelty_scores.csv")) %>%
  filter(!is.na(user_surprise) & !is.na(user2_surprise)) %>%
  mutate(
    user_resonance = user_surprise - user_transience,
    user2_resonance = user2_surprise - user2_transience,
    datatype = "Field"
```

```

) %>%
drop_na(user, user2) %>%
filter(interaction_count > 1, interaction_count < 10, user_surprise > -4, user2_surprise > -4)

```

1.2 1.2 Reshape Data

```

# Compute word counts for each agent's turn
df_res_field <- df_res_field %>%
  mutate(
    user_nwords = str_count(user, "\\S+"),
    user2_nwords = str_count(user2, "\\S+")
  )

df_res_long <- df_res_field %>%
  pivot_longer(
    cols = c(user_surprise, user_transience, user_nwords,
              user2_surprise, user2_transience, user2_nwords),
    names_to = c("agent", "metric"),
    names_pattern = "(user|user2)_(surprise|transience|nwords)"
  ) %>%
  pivot_wider(names_from = metric, values_from = value) %>%
  mutate(
    resonance = surprise - transience,
    agent = factor(agent, levels = c("user", "user2"), labels = c("User", "user2"))
  )

```

1.3 1.3 Descriptive Statistics

```

desc_stats <- df_res_long %>%
  group_by(agent) %>%
  summarise(
    n = n(),
    mean_surprise = mean(surprise, na.rm = TRUE),
    sd_surprise = sd(surprise, na.rm = TRUE),
    mean_transience = mean(transience, na.rm = TRUE),
    sd_transience = sd(transience, na.rm = TRUE),
    mean_resonance = mean(resonance, na.rm = TRUE),
    sd_resonance = sd(resonance, na.rm = TRUE)
  )

print(desc_stats)

## # A tibble: 2 x 8
##   agent      n mean_surprise sd_surprise mean_transience sd_transience
##   <fct> <int>         <dbl>         <dbl>         <dbl>         <dbl>
## 1 User    301         -1.68          0.630         -0.944          0.539
## 2 user2   301         -1.74          0.624         -0.918          0.593
## # i 2 more variables: mean_resonance <dbl>, sd_resonance <dbl>

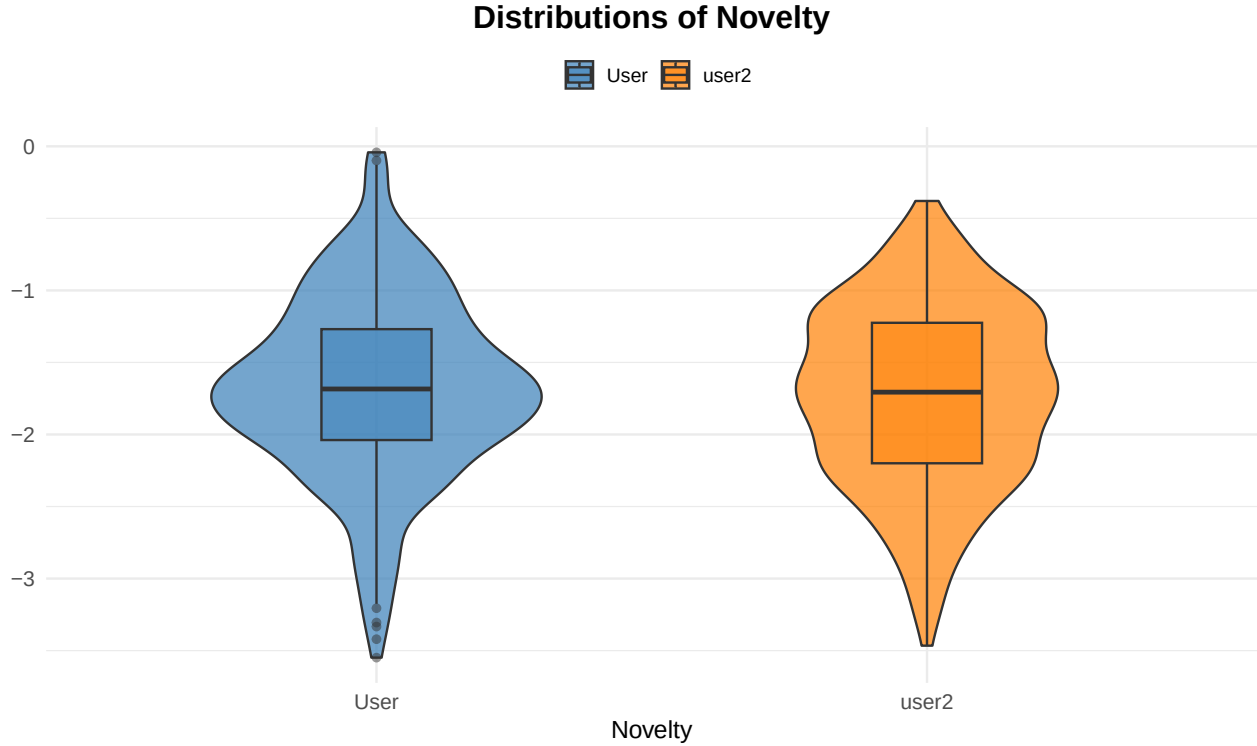
knitr::kable(desc_stats, caption = "Descriptive Statistics for Novelty Metrics by Agent", digits = 3)

```

Table 1: Descriptive Statistics for Novelty Metrics by Agent

agent	n	mean_surprise	sd_surprise	mean_transience	sd_transience	mean_resonance	sd_resonance
User	301	-1.680	0.630	-0.944	0.539	-0.737	0.762
user2	301	-1.737	0.624	-0.918	0.593	-0.819	0.739

```
p_novelty_agent <- ggplot(df_res_long, aes(x = agent, y = surprise, fill = agent)) +
  geom_violin(width = 0.6, alpha = 0.7) +
  geom_boxplot(width = 0.2, alpha = 0.5) +
  scale_fill_manual(values = c("User" = "#377EB8", "user2" = "#FF7F00"), name = "") +
  scale_color_manual(values = c("User" = "#377EB8", "user2" = "#FF7F00")) +
  labs(title = "Distributions of Novelty", x = "Novelty", y = NULL) +
  theme(
    legend.position = "top",
    plot.title = element_text(hjust = 0.5, size = 18, face = "bold"),
    axis.text = element_text(size = 12)
  )
print(p_novelty_agent)
```



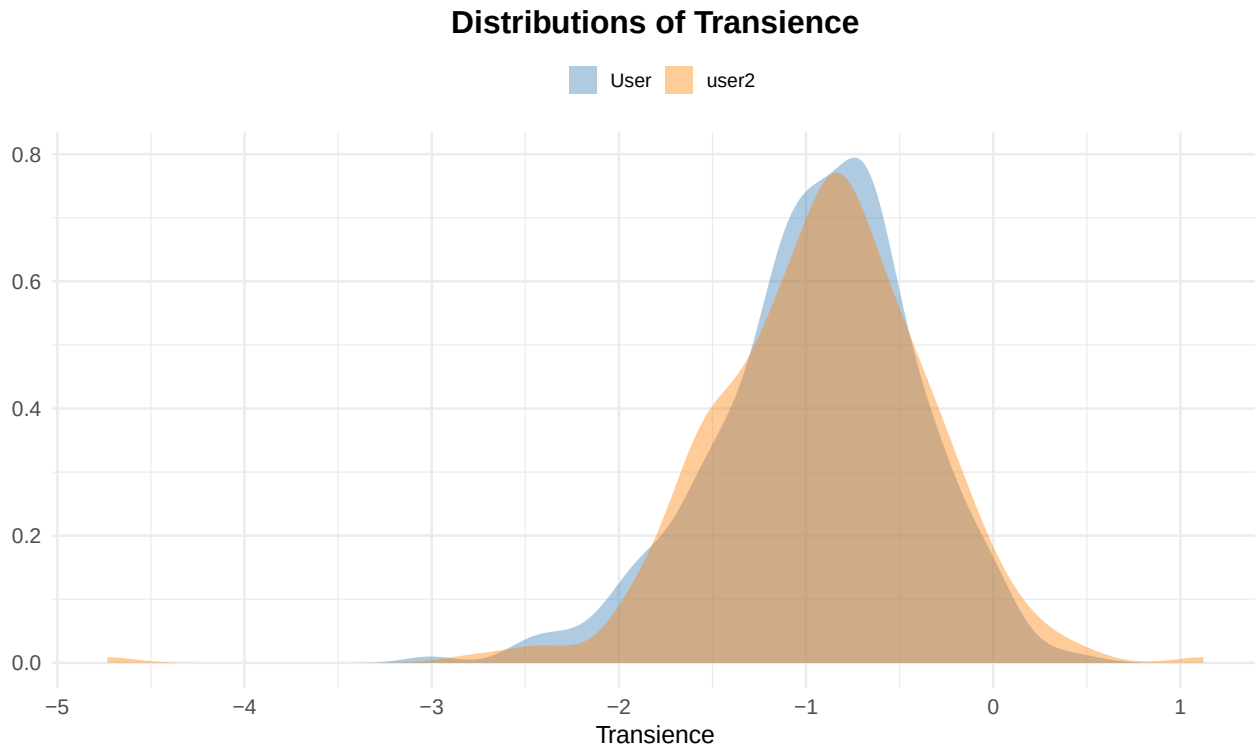
```
ggsave(
  file.path(GRAPHs_DIR, "Surprise_Distributions_Agent.png"),
  p_novelty_agent,
  width = 10,
  height = 6,
  dpi = 300
)

p_transience_agent <- ggplot(df_res_long, aes(x = transience, fill = agent)) +
  geom_density(alpha = 0.4, color = NA) +
```

```

scale_fill_manual(values = c("User" = "#377EB8", "user2" = "#FF7F00"), name = "") +
scale_color_manual(values = c("User" = "#377EB8", "user2" = "#FF7F00")) +
labs(title = "Distributions of Transience", x = "Transience", y = NULL) +
theme(
  legend.position = "top",
  plot.title = element_text(hjust = 0.5, size = 18, face = "bold"),
  axis.text = element_text(size = 12)
)
print(p_transience_agent)

```



```

ggsave(
  file.path(GRAPHS_DIR, "Transience_Distributions_Agent.png"),
  p_transience_agent,
  width = 10,
  height = 6,
  dpi = 300
)

```

2. Novelty and Resonance Analysis

2.1 Resonance vs Novelty Plot

```

# Plot Resonance vs Novelty (Field Data)
p_resonance <- ggplot(df_res_long, aes(x = surprise, y = resonance, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
  scale_color_manual(values = c("User" = "#377EB8", "user2" = "#FF7F00")) +
  geom_vline(data = desc_stats,
    aes(xintercept = mean_surprise, color = agent),

```

```

        linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
labs(
  title = "Resonance vs Novelty (Field Data)",
  x = "Novelty",
  y = "Resonance (Novelty - Transience)"
) +
theme_minimal(base_size = 14) +
theme(legend.position = "top")

print(p_resonance)

```



```

ggsave(file.path(GRAPHS_DIR, "novelty_resonance_vs_novelty.png"),
  p_resonance, width = 12, height = 6, dpi = 300)

```

2.2 Transience vs Novelty Plot

```

p_transience <- ggplot(df_res_long, aes(x = surprise, y = transience, color = agent)) +
  geom_point(alpha = 0.5, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE, fullrange = TRUE, size = 1.2) +
  scale_color_manual(values = c("User" = "#377EB8", "user2" = "#FF7F00")) +
  geom_vline(data = desc_stats,
    aes(xintercept = mean_surprise, color = agent),
    linetype = "dashed", linewidth = 0.8, show.legend = FALSE) +
labs(
  title = "Transience vs Novelty (Field Data)",
  x = "Novelty (Surprisal)",
  y = "Transience"
) +
theme_minimal(base_size = 14) +
theme(legend.position = "top")

print(p_transience)

```

Transience vs Novelty (Field Data)



```
ggsave(file.path(GRAPHS_DIR, "novelty_transience_vs_novelty.png"),
        p_transience, width = 12, height = 6, dpi = 300)
```

3. Statistical Models

3.1 Resonance Model

```
# Model without length control
model_resonance <- lmer(resonance ~ surprise * agent + (1 | conversation_id), data = df_res_long)
cat("=== Resonance Model ===\n")

## === Resonance Model ===

print(summary(model_resonance)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)    0.83995510 0.09429553 447.6859   8.907687 1.317103e-17
## surprise       0.93785477 0.04972026 591.6825  18.862629 1.710018e-62
## agentuser2     -0.16640281 0.12632693 578.7628  -1.317239 1.882796e-01
## surprise:agentuser2 -0.07932041 0.06956923 580.6523  -1.140165 2.546875e-01
cat("\n=== Slope Comparison (Resonance) ===\n")

##
## === Slope Comparison (Resonance) ===

print(emtrends(model_resonance, specs = "agent", var = "surprise"))

## agent surprise.trend      SE df lower.CL upper.CL
## User              0.938 0.0498 592   0.840   1.04
## user2             0.859 0.0514 598   0.757   0.96
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```

# Model WITH length control
model_resonance_len <- lmer(resonance ~ surprise * agent + nwords + (1 | conversation_id), data = df_res)
cat("\n=== Resonance Model (controlling for turn length) ===\n")

##
## === Resonance Model (controlling for turn length) ===
print(summary(model_resonance_len)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)    0.39063077 0.175057510 537.3305   2.231443 2.606352e-02
## surprise       0.91892214 0.049928016 585.7367  18.404940 4.895366e-60
## agentuser2     -0.19780855 0.125520032 573.6609  -1.575912 1.155973e-01
## nwords         0.01612153 0.005146423 565.0429   3.132570 1.822568e-03
## surprise:agentuser2 -0.09095036 0.068985968 575.2547  -1.318389 1.878980e-01
cat("\n=== Slope Comparison (Resonance, length-controlled) ===\n")

##
## === Slope Comparison (Resonance, length-controlled) ===
print(emtrends(model_resonance_len, specs = "agent", var = "surprise"))

## agent surprise.trend      SE df lower.CL upper.CL
## User          0.919 0.0500 587    0.821    1.017
## user2         0.828 0.0527 594    0.725    0.931
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

```

3.2 Transience Model

```

# Model without length control
model_transience <- lmer(transience ~ surprise * agent + (1 | conversation_id), data = df_res_long)
cat("=== Transience Model ===\n")

##
## === Transience Model ===
print(summary(model_transience)$coefficients)

##              Estimate Std. Error      df    t value      Pr(>|t|)
## (Intercept)   -0.83995510 0.09429553 447.6859  -8.907687 1.317103e-17
## surprise       0.06214523 0.04972026 591.6825   1.249898 2.118311e-01
## agentuser2     0.16640281 0.12632693 578.7628   1.317239 1.882796e-01
## surprise:agentuser2 0.07932041 0.06956923 580.6523   1.140165 2.546875e-01
cat("\n=== Slope Comparison (Transience) ===\n")

##
## === Slope Comparison (Transience) ===
print(emtrends(model_transience, specs = "agent", var = "surprise"))

## agent surprise.trend      SE df lower.CL upper.CL
## User          0.0621 0.0498 592   -0.0357    0.160
## user2         0.1415 0.0514 598    0.0404    0.243
##

```

```
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

# Model WITH length control
model_transience_len <- lmer(transience ~ surprise * agent + nwords + (1 | conversation_id), data = df_)

cat("\n=== Transience Model (controlling for turn length) ===\n")

##
## === Transience Model (controlling for turn length) ===
print(summary(model_transience_len)$coefficients)

##              Estimate Std. Error      df    t value    Pr(>|t|)
## (Intercept)   -0.39063077 0.175057510 537.3305 -2.231443 0.026063524
## surprise       0.08107786 0.049928016 585.7367  1.623895 0.104936197
## agentuser2     0.19780855 0.125520032 573.6609  1.575912 0.115597322
## nwords        -0.01612153 0.005146423 565.0429 -3.132570 0.001822568
## surprise:agentuser2 0.09095036 0.068985968 575.2547  1.318389 0.187898037

cat("\n=== Slope Comparison (Transience, length-controlled) ===\n")

##
## === Slope Comparison (Transience, length-controlled) ===
print(emtrends(model_transience_len, specs = "agent", var = "surprise"))

## agent surprise.trend      SE df lower.CL upper.CL
## User           0.0811 0.0500 587  -0.0171    0.179
## user2          0.1720 0.0527 594   0.0686    0.275
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

4 5. Session Information

```
sessionInfo()

## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.3
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
```



```
## other attached packages:
## [1] emmeans_2.0.1   here_1.0.2       afex_1.4-1       cowplot_1.1.3
## [5] patchwork_1.3.0 zoo_1.8-12       stringi_1.8.4    arrow_19.0.1.1
## [9] lmerTest_3.1-3  lme4_1.1-35.5    Matrix_1.7-1     lubridate_1.9.3
## [13] forcats_1.0.0   stringr_1.5.1    dplyr_1.1.4      purrr_1.0.2
## [17] readr_2.1.5     tidyr_1.3.1      tibble_3.2.1     ggplot2_4.0.2
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1   farver_2.1.2      S7_0.2.1
## [4] fastmap_1.2.0      TH.data_1.1-2     pacman_0.5.1
## [7] digest_0.6.37      timechange_0.3.0  estimability_1.5.1
## [10] lifecycle_1.0.5    survival_3.7-0    magrittr_2.0.3
## [13] compiler_4.4.2     rlang_1.1.7       tools_4.4.2
## [16] utf8_1.2.4         yaml_2.3.10       knitr_1.50
## [19] labeling_0.4.3     bit_4.0.5         plyr_1.8.9
## [22] RColorBrewer_1.1-3 abind_1.4-8       multcomp_1.4-26
## [25] withr_3.0.2        numDeriv_2016.8-1.1 grid_4.4.2
## [28] fansi_1.0.6        xtable_1.8-4      scales_1.4.0
## [31] MASS_7.3-61        tinytex_0.52      cli_3.6.5
## [34] mvtnorm_1.3-3      rmarkdown_2.28    crayon_1.5.3
## [37] ragg_1.3.2         generics_0.1.3    rstudioapi_0.16.0
## [40] reshape2_1.4.4     tzdb_0.4.0        minqa_1.2.8
## [43] splines_4.4.2      assertthat_0.2.1  parallel_4.4.2
## [46] vctrs_0.7.1        boot_1.3-31       sandwich_3.1-1
## [49] carData_3.0-5      car_3.1-3         hms_1.1.3
## [52] pbkrtest_0.5.3     bit64_4.0.5       Formula_1.2-5
## [55] systemfonts_1.1.0  glue_1.8.0        nloptr_2.1.1
## [58] codetools_0.2-20   gtable_0.3.6      pillar_1.9.0
## [61] htmltools_0.5.8.1  R6_2.6.1          textshaping_0.4.0
## [64] rprojroot_2.1.1    vroom_1.6.5       evaluate_1.0.3
## [67] lattice_0.22-6     backports_1.5.0   broom_1.0.6
## [70] Rcpp_1.0.13        coda_0.19-4.1     nlme_3.1-166
## [73] mgcv_1.9-1         xfun_0.54         pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.